

LESSON 5-7 • SESSION 2 • PRACTICE • FORM A

Surface Area of Triangular Prisms

Learning target – I can find the surface area of a triangular prism using its net. · 6.G.A.4

REMEMBER

- A triangular prism has **2 triangle ends** and **3 rectangles: 5 faces**.
- Triangle end: $\frac{1}{2} \times \text{base} \times \text{height}$. In a right triangle, the legs are base and height.
- Each rectangle: a triangle **side** $\times$ the prism's **length**.

WORKED EXAMPLE

Prism 6 ft long; ends: base 8 ft, height 3 ft, sides 5 ft.

Step 1 Ends: $2 \times (\frac{1}{2} \times 8 \times 3) = 24 \text{ ft}^2$.

Step 2 Rectangles: $(8 + 5 + 5) \times 6 = 108 \text{ ft}^2$.

Step 3 Add: $24 + 108 = 132$.

Answer SA = 132 ft²

WORDS triangular prism (*prisma triangular*) · lateral face (*cara lateral*) · surface area (*área superficial*)

PRACTICE

show your work · label units

1. Find each area.

a. 2 ends, base 6 ft, height 4 ft: _____ ft²

b. rectangle 5 ft by 10 ft: _____ ft²

c. 2 ends of 9 cm² each: _____ cm²

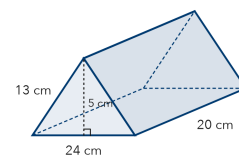
d. rectangles 3, 4, 5 cm by 11 cm: _____ cm²

2. **Candy Box.** Both slanted sides of this candy box are 13 cm.

a. Area of both triangle ends _____ cm²

b. Area of the three rectangles _____ cm²

c. Surface area _____ cm²



Show your work.

3. **Ramp Wedge.** A wedge is a triangular prism 14 cm wide. Its ends are right triangles with sides 20, 21, and 29 cm. Find its surface area.

Show your work.

Answer _____ cm²

4. **Cheese Wedge.** A cheese wedge is a prism 6 cm thick. Its ends are right triangles with sides 5, 12, 13 cm. Find its surface area.

Hint The 6 cm is one side of every rectangle.

Show your work.

Answer _____ cm²

5. **Nolan's Ends.** A prism is 22 in long. Each end has base 14 in and height 24 in. Nolan says both ends cover 168 in². Is he right?

Hint How many triangle ends does a prism have?

Show your work.

Answer _____ in²

6. **Roof Piece.** A roof piece is a prism 25 ft long. Each end: base 36 ft, height 24 ft, two 30 ft sides. Find its SA.

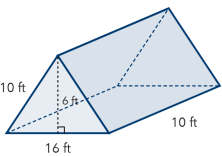
Show your work.

Answer _____ ft²

7. **Planter Wedge.** Find the surface area of the planter.

Hint Height for the triangles, sides for the rectangles.

Show your work.



Answer _____ ft^2

8. **Camping Tent.** A tent is a triangular prism 10 ft long. Each end: base 8 ft, height 3 ft, two 5 ft sides. The floor is 8 ft by 10 ft.

- a. Area of the two ends _____ ft^2
- b. Total surface area _____ ft^2
- c. Fabric needed with no floor _____ ft^2

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Amara’s Work.** A prism is 10 cm long. Its ends are right triangles with sides 6, 8, 10 cm. Amara found the rectangles (240 cm^2) and one end (24 cm^2) and wrote “ $\text{SA} = 264\text{ cm}^2$.” Explain and fix it.

TEST PRACTICE

choose the best answer

10. How many rectangular faces does a triangular prism have?

- A** 2
- B** 3
- C** 5
- D** 6

11. Each rectangle in the net uses one triangle side times the –

- A** triangle’s height
- B** prism’s length
- C** triangle’s area
- D** number of faces

SELF-CHECK CODE

find your answer · copy its letter into that problem’s box · spell a math word

3,264 N	1,400 R	336 A	1,120 T	124 E	800 Z	168 Q	240 I	204 L	456 G
#2c	#3	#4	#5	#6	#7	#8b	#8c		

CHECK YOURSELF

☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #____